

TruthLens AR: Bangla Multimodal Fake News, Image & Deepfake Detection Using Vision-Language Models

Multimodal Fake-News Detection with Transformer Text Encoders, CNN and Transformer Vision Encoders, and Weighted Late Fusion: An Empirical Study with a Reproducibility Audit on Fakeddit

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CSE 499A Senior Design Capstone | Repository: github.com/joy-saha-ayan/CSE499-Project

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Abstract—Multimodal fake-news detection is usually reported as a single headline accuracy. This paper takes the opposite approach. We re-derive every number in a complete capstone system directly from its executed notebooks, and we report both what the system achieves and where its measurements do not support the claims made about them. On a 17 045-sample multimodal subset of Fakeddit we fine-tune and evaluate four textual classifiers (a TF-IDF plus logistic-regression baseline, BERT-base, multilingual BERT, and XLM-RoBERTa-base) and three visual classifiers (ResNet-50, ViT-B/16, and EfficientNet-B0) under a single stratified 80/20 partition, and combine the deployed text and image branches with weighted late fusion. BERT-base attains the highest test accuracy at 0.8566, followed by mBERT at 0.8542 and XLM-RoBERTa at 0.8483; among the visual models EfficientNet-B0 leads at 0.8208, ahead of ResNet-50 at 0.7870 and ViT-B/16 at 0.7580. A second, independent experiment using frozen sentence embeddings with a ResNet-152 vision branch and intermediate feature-level fusion reaches 0.7020 mean batch accuracy on a smaller sample. The audit component of this work establishes four findings that a conventional write-up would have concealed: the fusion evaluation was executed with a randomly initialised text head, so the reported multimodal figure of 0.7891 measures the image branch alone; the fusion weight used offline (0.50) differs from the one deployed (0.60), and the deployed setting has never been evaluated; the deployed application inverts the Fakeddit label convention; and no validation partition exists, so the test split doubles as the monitoring set. We therefore conclude that this study demonstrates competitive unimodal performance and a working end-to-end deployment, but does not yet demonstrate a multimodal gain. We specify the minimal set of corrective experiments required, and outline a research roadmap towards Bangla multimodal misinformation detection, for which the binding constraint is corpus construction rather than model architecture.

Index Terms—Fake-news detection, multimodal learning, late fusion, transformer models, XLM-RoBERTa, convolutional neural networks, Fakeddit, reproducibility, low-resource languages, Bangla natural language processing.

I. INTRODUCTION

ONLINE misinformation is no longer a purely textual phenomenon. A misleading claim now travels as a package: a short, emotionally charged headline bound to an image that is manipulated, recycled from an unrelated event, or simply

captioned dishonestly. Large-scale measurement on Twitter found that false stories diffuse significantly farther, faster and more broadly than true ones, and that this gap is driven by human sharing behaviour rather than by automation [1]. Detection systems that inspect only the text of such an item therefore discard evidence that human fact-checkers routinely use.

The research community responded with multimodal detectors that encode text and image jointly [13]– [15], and with benchmark corpora large enough to train them. Fakeddit [3] is the most widely used of these: it contains more than one million Reddit submissions with paired text and images, labelled at two, three and six levels of granularity through distant supervision over curated subreddits. Its scale and its explicit multimodal design make it the natural starting point for an undergraduate capstone in this area.

This paper reports such a project. It has two purposes, and the second is unusual enough to state plainly at the outset.

The first purpose is conventional. We benchmark seven neural and statistical classifiers, four textual and three visual, on an identical 17 045-sample subset of Fakeddit under an identical stratified partition, and we combine the two deployed branches with weighted late fusion behind a Streamlit interface. This yields a controlled comparison of encoder families under a fixed data and optimisation budget, which is a legitimate contribution in its own right: most published comparisons vary the corpus, the split, or the training schedule simultaneously with the architecture.

The second purpose is a *reproducibility audit*. Every metric in this paper was recovered from the stored output cells of the executed notebooks in the project repository, not from any prose summary of them. Doing so revealed that the repository’s own documentation reports accuracies that its notebooks do not produce, and—more consequentially—that the multimodal experiment on which the project’s headline claim rests was executed with an untrained text encoder. We report this rather than quietly substituting a more favourable number, because a fusion result obtained from a randomly

initialised classification head is not a weak result; it is a measurement of something other than the intended system.

A. Contributions

The specific contributions of this work are:

- A controlled seven-model comparison across two modalities on a fixed Fakeddit subset, with all preprocessing, optimisation and evaluation settings held constant within each modality (Sections IV, V and IX).
- A precise description of two distinct fusion designs actually present in the project—probability-level late fusion and feature-level intermediate fusion—together with the exact parameterisation of each (Section VI).
- A verification report that cross-checks every published figure against the notebook cell that produced it, isolates four defects, and quantifies the effect of each on the conclusions that may be drawn (Section XII and Table VII).
- An explicit statement of what the project does *not* establish, and the minimal corrective experiment set that would establish it (Section XIV).
- A research roadmap for extending this English system to Bangla, in which the principal obstacle is identified as the absence of any paired Bangla text-image misinformation corpus rather than any modelling gap (Section XIII).

B. Organisation

Section II reviews related work. Section III describes the corpus and formalises the task, including a careful treatment of label semantics that turns out to matter. Sections IV and V present the textual and visual branches. Section VI develops the fusion formulation. Sections VII and VIII give the experimental setup and results, Section IX the cross-model comparison and Section X the error analysis. Section XI describes deployment, Section XII the limitations and the verification report, Section XIII the Bangla roadmap, and Sections XIV and XV the discussion and conclusion.

II. BACKGROUND AND RELATED WORK

A. Textual fake-news detection

Early automated detectors treated veracity as a text-classification problem over hand-engineered lexical, syntactic and psycholinguistic features [2]. Bag-of-words representations weighted by term frequency-inverse document frequency, fed to a linear classifier, remain a strong and extremely cheap baseline, and we retain one in this study for exactly that reason.

The transformer architecture [4] and the masked-language-modelling objective of BERT [5] displaced these methods. BERT produces contextual token representations from a bidirectional encoder pre-trained on unlabelled corpora, after which a task head can be fine-tuned with comparatively little labelled data. RoBERTa [6] showed that BERT was substantially under-trained and that removing the next-sentence objective while enlarging the batch and corpus improves downstream accuracy. Multilingual BERT extends the same recipe to 104 languages using a single shared WordPiece vocabulary, and exhibits surprisingly strong zero-shot cross-lingual transfer despite having no explicit cross-lingual signal [8]. XLM-RoBERTa [7] combines the RoBERTa training

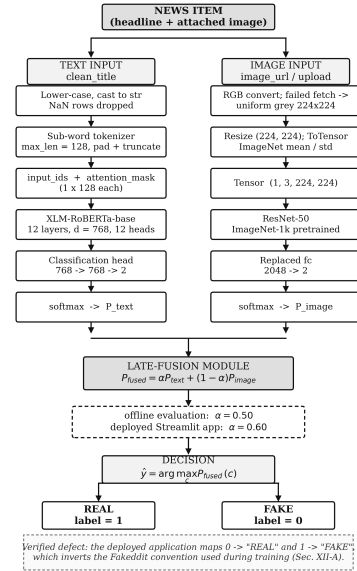


Fig. 1. End-to-end architecture of the deployed multimodal system as it exists in the repository. The two branches are trained separately and never exchange gradients; the fusion module has no learnable parameters. The dashed annotation records a defect confirmed by source inspection and discussed in Section XII-A.

regime with a 250 000-token SentencePiece vocabulary trained on 2.5 TB of filtered CommonCrawl covering 100 languages, and outperforms mBERT on most cross-lingual benchmarks. All three are evaluated here.

B. Visual and multimodal detection

On the visual side, residual learning [9] made very deep convolutional networks trainable by introducing identity shortcuts that let gradients bypass stacked non-linearities; ResNet-50 remains the default ImageNet [12] backbone for transfer learning. EfficientNet [11] instead searches for a balanced compound scaling of depth, width and resolution, reaching comparable accuracy with roughly one fifth of the parameters. The Vision Transformer [10] abandons convolution entirely, treating 16×16 image patches as tokens for a standard transformer encoder; it excels at very large pre-training scale but is comparatively data-hungry when fine-tuned on small task-specific corpora, a property that our results reproduce.

Multimodal detectors combine these families. EANN [13] adds an adversarial event discriminator so that the shared representation transfers to unseen news events rather than memorising event-specific cues. MVAE [14] learns a joint latent space with a variational autoencoder whose reconstruction objective regularises the detector. SpotFake [15] takes the simplest effective route, concatenating frozen BERT and VGG-19 features before a small classifier, and still outperforms both predecessors on Twitter and Weibo data. SAFE [30] and co-attention models such as MCAN [31] explicitly measure text-image agreement rather than merely pooling the two streams. Baltrušaitis *et al.* [16] provide the standard taxonomy of fusion strategies that we adopt in Section VI.

C. Fakeddit and its known pitfalls

Fakeddit [3] labels items by the subreddit from which they were harvested, a form of distant supervision [33]. This

yields a million labelled multimodal samples at negligible annotation cost, but it also means that community-specific style is correlated with the label. A model can therefore reach high accuracy by recognising the visual and lexical register of, say, *r/photshopbattles* versus *r/nottheonion* without acquiring any notion of veracity—precisely the shortcut-learning failure mode catalogued by Geirhos *et al.* [28]. VERITE [32] documents a related unimodal-bias problem in multimodal misinformation benchmarks, where a detector can exploit one modality’s artefacts and appear multimodal without being so. Segura-Bedmar and Alonso-Bartolomé [34] report on Fakeddit that a text-image CNN reaches about 87% on the six-way task while text-only BERT reaches about 78%, and note that manipulated-content and false-connection categories benefit most from the image. These caveats frame our own results and are revisited in Section X.

D. Bangla and low-resource misinformation

Bangla is spoken by more than 200 million people yet is heavily under-served by misinformation research. BanFake-News [20] is the principal resource: approximately 50 000 annotated Bangla news articles with benchmark systems built from linguistic features and neural encoders. It is, however, *text-only*. BanglaBERT [21] pre-trains an ELECTRA-style discriminator on 27.5 GB of Bangla text and outperforms both mBERT and XLM-R (large) on Bangla natural-language understanding while remaining far cheaper to fine-tune. IndicBERT [22] offers an alternative multilingual encoder covering major Indic languages. To the best of our knowledge, no publicly available Bangla corpus pairs news text with the accompanying image and a veracity label, which is the central premise of the roadmap in Section XIII.

E. Positioning of this work

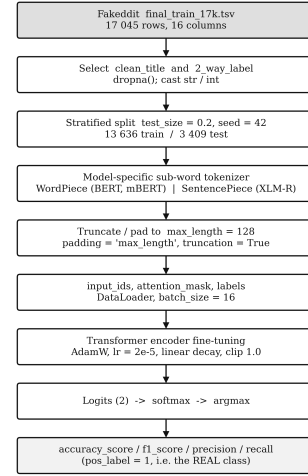
We do not claim a new architecture. Our contribution is a controlled comparison under a fixed budget, plus an audit methodology that we believe deserves wider use. The literature comparison in Table VIII places our verified numbers beside published Fakeddit results and makes clear which comparisons are legitimate and which are not.

III. DATASET AND PROBLEM FORMULATION

A. The working subset

All primary experiments use *final_train_17k.tsv*, a 17 045-row extract of the Fakeddit multimodal-only release stored in the project repository. The file has sixteen columns, of which four are used: *clean_title* (a lower-cased, punctuation-stripped headline), *image_url* (a Reddit preview CDN link), *2_way_label* (the binary target) and *hasImage*, which is true for all 17 045 rows. Neither *clean_title* nor *image_url* contains a null value, so the *dropna* calls in the preprocessing code remove nothing and the textual and visual partitions remain index-aligned. This alignment is not incidental: it is what makes row-wise late fusion valid in Section VI, and it was verified rather than assumed.

Headlines are short. The mean length is 42.1 characters with a standard deviation of 32.1, a median of 34 and a maximum of 470. A 128-token cap therefore truncates almost nothing, and the entire task is closer to short-text classification than to



No validation partition is created; the 3 409-sample test split is also used for per-epoch monitoring (Sec. XII-B).

Fig. 2. Complete textual processing pipeline shared by all four text models. Only the tokenizer and the encoder differ between models; the corpus, the split, the sequence length, the batch size and the optimiser schedule are held constant, which is what makes the comparison in Table III interpretable.

document classification. This has an important consequence for interpretation, developed in Section X: the textual models are working from a headline alone, with no article body, source or publication date.

B. Label semantics

The polarity of *2_way_label* is the single most consequential detail in this dataset, and it is easy to get backwards. We resolved it by cross-tabulating the binary label against the six-way label and against the source subreddit. The result is unambiguous: every row with *2_way_label* = 1 has *6_way_label* = 0, which is Fakeddit’s *true* category, and these rows originate from *r/mildlyinteresting*, *r/photshopbattles*, *r/nottheonion*, *r/upliftingnews*, *r/neutralnews* and *r/usnews*. Every row with *2_way_label* = 0 carries a six-way label between 1 and 5—satire, false connection, imposter content, manipulated content or misleading content—and originates from *r/psbattl_artwork*, *r/pareidolia*, *r/fakehistoryporn*, *r/fakealbumcovers*, *r/misleadingthumbnails* and *r/propagandaposters*. Table II records the resulting composition.

Therefore **label 1 denotes REAL** and **label 0 denotes FAKE**. Two consequences follow. First, because *scikit-learn* [25] defaults to *pos_label* = 1, every precision, recall and F1 value reported in this paper characterises retrieval of *real* news, not detection of fake news; we state this explicitly wherever such a number appears. Second, the deployed application maps label 0 to the string “REAL NEWS”, which inverts the convention and is analysed as defect D3 in Section XII-A.

C. Class balance and the honest baseline

The subset contains 10 404 fake items (61.04%) and 6 641 real items (38.96%). A classifier that predicts the majority class unconditionally therefore attains 61.04% accuracy and 0.0 F1 for the positive class. This, not 50%, is the floor against which every accuracy in Section VIII must be read, and it is

TABLE I
REPOSITORY INVENTORY AND PROVENANCE

Artefact	Role in this study	Origin
Fake_News_Detection/ CSE499A_Final.ipynb	Primary experiment: all seven models plus fusion	Own work
499A_FakeNews/ CSE499A.ipynb	XLm-R fine-tuning via HF Trainer; loss 0.4460 over 2 559 steps	Own work
Fake_News_Detection/ app.py	Deployed Streamlit application (459 lines)	Own work
others/Review_3.ipynb	Secondary experiment: intermediate fusion runs	Own work
others/TruthLens_ XLM_RoBERTa.ipynb	Text baseline; contains no executed output	Own work
integrating_model/	Joint-fusion codebase driven by Review_3	Third party [35]
Fakeddit-WebApp/	Six-way BERT+ResNet-50 reference notebook; not used for any claim here	Third party
review_2/	EfficientNet-B0 video deepfake detector; unrelated task	Third party

Provenance was established from commit history and from attribution statements inside each sub-project. Only artefacts marked as own work contribute evidence to any claim in this paper.

drawn as a reference line in Fig. 18(a). No class weighting, resampling or focal loss was applied in any experiment; the loss is unweighted cross-entropy throughout.

D. Task formulation

Let a news item be a pair $x = (t, v)$ where t is a headline string and v an image, with label $y \in \{0, 1\}$. A textual encoder f_T and a visual encoder f_V each produce class logits, which softmax converts to distributions:

$$P_{\text{text}}(c | t) = \exp(z_{t,c}) / \sum_{c'} \exp(z_{t,c'}) \quad (1)$$

and analogously for P_{image} . The unimodal experiments in Sections IV and V evaluate $\arg \max_c$ of each distribution separately; the multimodal experiment in Section VI combines them.

E. Partitioning

A single stratified split is used throughout: `train_test_split(test_size=0.2, random_state=42, stratify=label)`, giving 13 636 training and 3 409 evaluation samples with the class ratio preserved in both. The fixed seed makes the split reproducible, and identical seeds and inputs are used for the textual and visual partitions, which is what preserves row alignment.

There is no validation partition. The 3 409-sample split is used both for per-epoch monitoring during training and as the final reported test set. This is defect D5 of Section XII. Its practical effect here is bounded—no early stopping, checkpoint selection or hyper-parameter search is driven by the monitored metric, and the final-epoch weights are simply kept—but the reported accuracies are nonetheless optimistic in the strict sense, because the experimenter observed test-set performance before fixing the number of epochs.

IV. TEXTUAL MODELING

A. Text preprocessing

Preprocessing is deliberately minimal, and is identical for all four textual models. The `clean_title` field is cast to string and the label to integer; no stemming, stop-word removal, spell correction or emoji handling is applied. Each

TABLE II
COMPOSITION AND LABEL SEMANTICS OF THE WORKING SUBSET

Binary label	6-way label	Meaning	n	Share
1	0	True content (REAL)	6 641	38.96 %
0	1	Satire / parody	1 013	5.94 %
0	2	False connection	3 134	18.39 %
0	3	Imposter content	332	1.95 %
0	4	Manipulated content	5 248	30.79 %
0	5	Misleading content	677	3.97 %
0	1–5	All fake categories	10 404	61.04 %
		Total	17 045	100 %

Derived by cross-tabulating the two label columns and the source subreddit over all 17 045 rows. The majority class is FAKE, so the trivial baseline accuracy is 0.6104.

headline is passed to the tokeniser bundled with its own checkpoint, with `max_length=128`, `padding='max_length'` and `truncation=True`, producing an `input_ids` tensor and an `attention_mask` tensor of shape 1×128 per sample. Batches of sixteen are drawn with shuffling for training and without for evaluation. Fig. 2 shows the complete flow.

The uniformity is the point. Because the corpus, the split, the sequence length, the batch size and the optimiser schedule are identical, differences in Table III between the transformer models are attributable to the encoder and its tokeniser, not to data handling.

B. TF-IDF with logistic regression

The baseline vectorises the training headlines with `TfidfVectorizer(max_features=5000, ngram_range=(1,2))` and fits `LogisticRegression(max_iter=1000, random_state=42)`. It has no pre-training, no contextual representation and no notion of word order beyond bigrams. Its role is to quantify how much of the task is solvable by surface lexical cues alone. Its accuracy of 0.7715 is 16.1 points above the majority baseline, which tells us that a substantial fraction of the signal in this corpus is shallow and lexical—a finding we return to in Section X.

C. BERT-base-uncased

BERT [5] was selected as the English-specific upper reference. The checkpoint is `bert-base-uncased`: twelve encoder layers, hidden size 768, twelve attention heads, feed-forward width 3072 with GELU activation, dropout 0.1, a 30 522-entry WordPiece vocabulary and 110 M parameters. Because the corpus is already lower-cased, the uncased variant is the natural match. Fig. 3 traces the full path from raw text to class probability.

The classification head is the one supplied by `AutoModelForSequenceClassification`: the pooled [CLS] state passes through a $768 \rightarrow 768$ dense layer with tanh, dropout, and a $768 \rightarrow 2$ output projection. It is randomly initialised, which the library reports as a warning; that warning is expected here and should not be confused with the genuinely problematic instance discussed in Section VIII-C.

Training uses AdamW [23] at a learning rate of 2×10^{-5} with a linear decay schedule, zero warm-up, gradient-norm clipping at 1.0, and two epochs. BERT reaches 0.8530 accuracy after one epoch and 0.8566 after two—the highest of any model in the study.

D. Multilingual BERT

The checkpoint `bert-base-multilingual-uncased` shares BERT’s twelve-layer, 768-dimensional topology but replaces

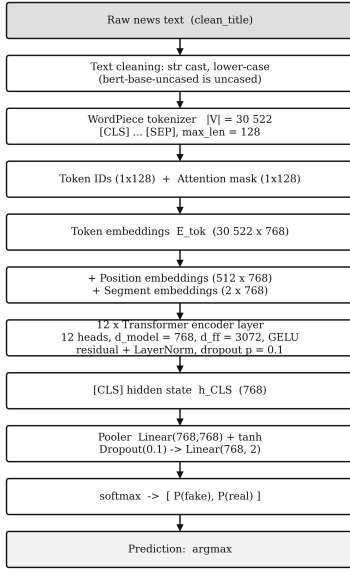
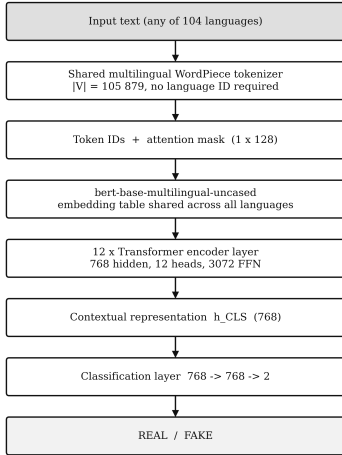


Fig. 3. BERT-base-uncased classification branch. Dimensions are those of the actual *bert-base-uncased* checkpoint used. Segment embeddings are present in the architecture but carry no information here, since every input is a single unpaired headline.



A single shared sub-word vocabulary lets one parameter set serve English training and, in principle, Bangla inference (Sec. XIII).

Fig. 4. Multilingual BERT branch. The architecture is identical to Fig. 3 apart from the shared 105 879-token vocabulary. The relevance of this design to Bangla misinformation detection is discussed in Section XIII; no Bangla experiment was performed in the present study.

the vocabulary with a 105 879-entry WordPiece inventory learned jointly over Wikipedia in 104 languages, giving roughly 168 M parameters, of which the embedding table dominates. Crucially, no language identifier is supplied at either training or inference time: a single parameter set serves all languages, which is what makes zero-shot cross-lingual transfer possible [8]. Fig. 4 gives the pipeline.

mBERT is included specifically because the project’s stated long-term goal is Bangla misinformation detection. Bangla is among the 104 pre-training languages, so mBERT trained on English Fakeddit can in principle be applied to Bangla input without retraining. That capability is *not* tested in this work—

no Bangla evaluation was run—and it is listed as future work in Section XIII, not as a result. Under matched conditions mBERT scores 0.8542, only 0.24 points below monolingual BERT, which is a small price for 104-language coverage.

E. XLM-RoBERTa

XLM-RoBERTa-base [7] is the model the project ultimately deployed. Its configuration was read directly from the printed module summary in the repository’s notebook rather than from documentation: twelve *XLMRobertaLayer* blocks; hidden size 768; twelve heads with query, key and value projections of 768×768 ; an intermediate dense layer of $768 \rightarrow 3072$ with GELU; layer normalisation with $\varepsilon = 1 \times 10^{-5}$; dropout 0.1 at attention, output and embedding positions; a position-embedding table of 514×768 ; and a SentencePiece vocabulary of 250 002 entries. Total parameters are approximately 278 M, of which roughly 192 M reside in the embedding table alone. Fig. 5 renders this in full.

XLM-R differs from mBERT in three respects that matter: it uses SentencePiece rather than WordPiece, which handles Bangla’s conjunct consonants and non-space-delimited morphology more gracefully; it was trained on CommonCrawl rather than Wikipedia, which is far larger and more colloquial; and it drops the next-sentence-prediction objective in favour of the RoBERTa recipe [6].

Fine-tuning used AdamW at 2×10^{-5} for three epochs, one more than the other transformers. The trajectory is 0.8187, 0.8354, 0.8483, still rising at the final epoch, so the model is likely under-trained rather than over-fitted. An independent run using the Hugging Face *Trainer* API on the same split converged to a training loss of 0.4460 over 2 559 optimiser steps in 1 202.8 s; its evaluation metrics were rendered as an interactive widget and are consequently not recoverable from the stored notebook, so they are recorded as *not available in the current project artefacts*.

The fine-tuned weights were exported to *xlm_roberta_trained.pt*, a 1.11 GB state dictionary containing 201 tensors. A reload test produced logits of $[2.341, -1.967]$ on a probe string, confirming that the file holds trained rather than freshly initialised parameters. That confirmation matters, because the fusion experiment failed to use this file at all.

F. Text branch selection

Fig. 6 summarises the comparison and the decision taken. BERT-base is the most accurate text model at 0.8566, yet XLM-RoBERTa (0.8483) was the one exported and deployed. The 0.83-point deficit was accepted in exchange for multilingual capability aligned with the project’s Bangla roadmap. This is a defensible engineering decision, but it should be recorded as what it is: a deliberate trade of accuracy for future extensibility, not a claim that XLM-R is the best model on this corpus. No statistical test was performed to determine whether the 0.83-point gap is significant; with 3 409 test samples it corresponds to about 28 items, and establishing significance would require either bootstrap resampling or a McNemar test on paired predictions. Neither was run, and we therefore treat BERT-base and XLM-R as empirically indistinguishable on the present evidence.

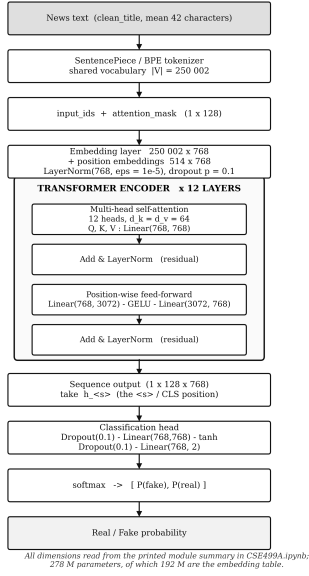
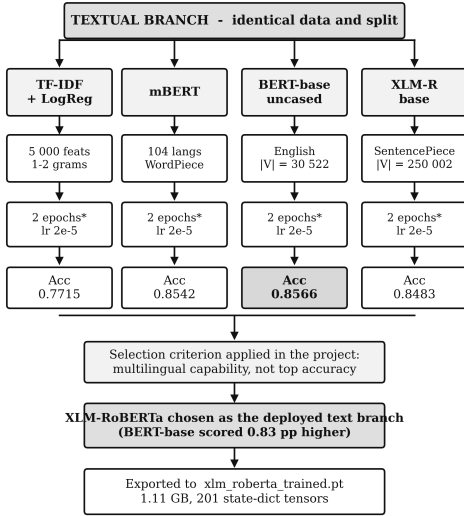


Fig. 5. XLM-RoBERTa-base branch, the deployed text encoder. Every dimension shown was read from the printed *XLMRobertaModel* summary in the project notebooks; none is assumed from documentation. As shown, the encoder stack is repeated twelve times, and only the representation at the sequence-initial position is passed to the classification head.



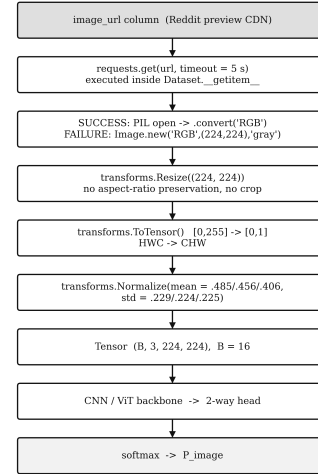
* XLM-RoBERTa was trained for 3 epochs; all other transformers for 2. Reported accuracy is on the 3 409-sample test split.

Fig. 6. Text-model comparison pipeline and the deployment decision it led to. All four models see identical inputs and splits, so the accuracies are directly comparable. The selected model is not the most accurate one; the rationale is stated in Section IV-F.

V. VISUAL MODELING

A. Image preprocessing and its principal weakness

Images are not stored locally. They are fetched over HTTP from the *image_url* column inside *Dataset.__getitem__*, with a five-second timeout, once per sample per epoch. Successful responses are decoded with PIL and converted to RGB. The transform chain is *Resize((224,224))*, *ToTensor()* and *Normalize* with the standard ImageNet statistics (mean 0.485/0.456/0.406, standard deviation 0.229/0.224/0.225). No augmentation of any kind is applied, and the resize does not preserve aspect ratio. Fig. 7 shows the complete path.



No augmentation is applied. Because a failed download yields a constant grey tensor that keeps its original label, download failures inject label noise into both the training and the test split (Sec. XII-C).

Fig. 7. Visual processing pipeline shared by ResNet-50, ViT-B/16 and EfficientNet-B0. The dashed observation records the grey-substitution failure path, which is the dominant threat to the validity of every visual number in this paper (Sections V-A and XII-C).

The failure branch is the problem. When a request times out or returns a non-image, the exception handler substitutes a uniform grey 224×224 image—while retaining the sample’s original label. Every such sample becomes a constant tensor carrying an arbitrary target, in both the training and the evaluation partition. In training this is label noise; in evaluation it is worse, because a deterministic constant input paired with a 61/39 label distribution systematically inflates apparent accuracy toward the majority rate. The failure rate on the 17 045-sample run was never logged and is therefore recorded as *requires additional experiment*. However, the secondary experiment of Section VIII-D *did* log it on a comparable Fakeddit sample: 1 679 of 5 000 URLs failed, a rate of 33.6 %. If the rate in the primary experiment is even of that order, the visual accuracies in Table III are not trustworthy at the reported precision.

A related consequence deserves separate emphasis: because the fetch is repeated every epoch, the set of failed samples is not stable across epochs. The effective training corpus therefore changes between epoch 1 and epoch 2 in an uncontrolled way, and the visual results are not exactly reproducible even with a fixed seed.

B. ResNet-50

ResNet-50 [9] is instantiated from *torchvision.models.resnet50(pretrained=True)* with ImageNet-1k weights, and its 1000-way classifier is replaced by *nn.Linear(2048, 2)*. The stem applies a 7×7 convolution with 64 filters at stride 2 followed by batch normalisation, ReLU and a 3×3 max-pool at stride 2. Four residual stages then contain three, four, six and three bottleneck blocks respectively, each block being a 1×1 reduction, a 3×3 convolution and a 1×1 expansion with an identity shortcut summed before the final ReLU. Spatial resolution falls $224 \rightarrow 112 \rightarrow 56 \rightarrow 28 \rightarrow 14 \rightarrow 7$ while channel depth rises

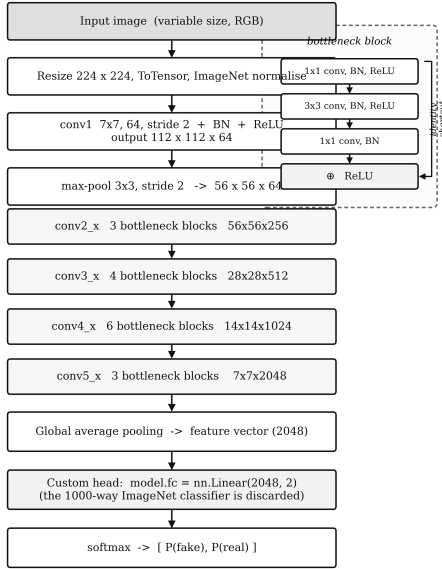


Fig. 8. ResNet-50 branch with the project’s custom two-way head. The inset expands one bottleneck block and shows the identity shortcut that carries the gradient past the stacked convolutions; this is the mechanism that makes fifty-layer training stable.

to 2048; global average pooling yields the 2048-dimensional feature that the new head consumes. Fig. 8 shows both the macro-architecture and the bottleneck detail.

Training used Adam [24] at 1×10^{-4} with unweighted cross-entropy for two epochs, no gradient clipping and no learning-rate schedule. The result was 0.7870 accuracy with recall 0.8840 but precision only 0.6724. The trained head was exported as a 94 MB, 320-tensor state dictionary and verified to reload with logits [6.766, −5.181].

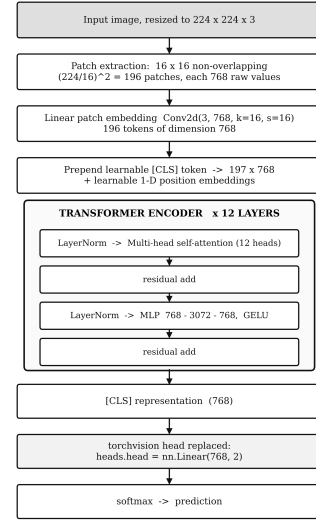
C. Vision Transformer

ViT-B/16 [10] is loaded via *torchvision.models.vit_b_16* (checkpoint *vit_b_16-c867db91.pth*, 330 MB) and its head replaced with a two-way linear layer. The 224×224 input is partitioned into 196 non-overlapping 16×16 patches, each projected to 768 dimensions by a strided convolution; a learnable class token is prepended and learnable position embeddings are added, giving a 197×768 sequence for twelve pre-norm encoder layers with twelve heads and a 3072-wide MLP. Fig. 9 shows the pipeline.

ViT is the weakest visual model here at 0.7580, and it is the only model whose accuracy *fell* between epochs, from 0.7688 to 0.7580, with F1 dropping more sharply from 0.7372 to 0.6939. This is consistent with the known data-appetite of vision transformers: lacking the locality and translation-equivariance priors built into convolution, ViT must learn them from data, and 13 636 noisy images across two epochs is far too little. It is also consistent with over-fitting under a learning rate of 1×10^{-4} , which is aggressive for transformer fine-tuning; a rate an order of magnitude smaller would be the standard choice. We cannot separate these two explanations from the available evidence.

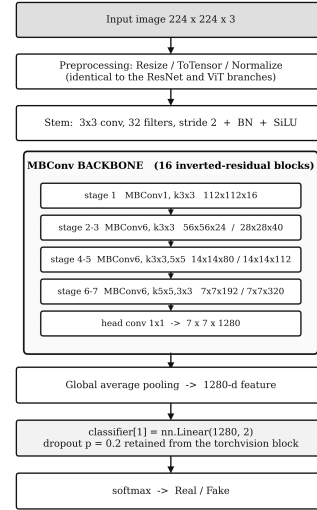
D. EfficientNet-B0

EfficientNet-B0 [11] is loaded from torchvision with the *efficientnet_b0_rwrightman* weights (20.5 MB) and its *classifier[1]*



Instantiated as *torchvision.models.vit_b_16(pretrained=True)*; checkpoint *vit_b_16-c867db91.pth* (330 MB) downloaded at run time.

Fig. 9. ViT-B/16 branch. Patch tokenisation replaces convolutional feature extraction entirely. As discussed in Section V-C, this model was the only one whose test accuracy declined between epochs.

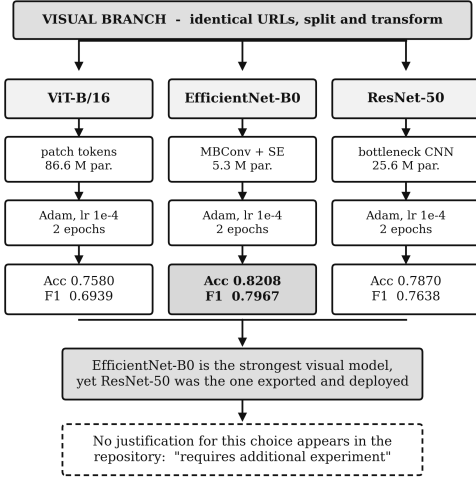


Each MBConv block contains a depth-wise separable convolution and a squeeze-and-excitation branch; only the final classifier was replaced.

Fig. 10. EfficientNet-B0 branch. Compound scaling and inverted bottlenecks deliver the best visual accuracy in this study at roughly one fifth of ResNet-50’s parameter count.

replaced by a $1280 \rightarrow 2$ linear layer. The backbone comprises a strided 3×3 stem followed by sixteen mobile inverted-bottleneck blocks with depth-wise separable convolutions, squeeze-and-excitation attention and SiLU activation, arranged in seven stages, then a 1×1 head convolution to 1280 channels and global average pooling. Fig. 10 gives the layout.

EfficientNet-B0 is the strongest visual model at 0.8208 accuracy and 0.7967 F1, beating ResNet-50 by 3.4 accuracy points while using roughly one fifth of the parameters (5.3 M against 25.6 M). It also achieves the highest recall of any model in the entire study at 0.9014.



All three share the same failure mode: recall for the REAL class is high (0.70-0.90) while precision is low (0.67-0.71), i.e. the visual models over-predict REAL on a corpus where 61 % of items are FAKE.

Fig. 11. Image-model comparison. The strongest visual model is not the deployed one; the discrepancy is documented rather than resolved, because the repository contains no evidence bearing on the decision.

E. Visual branch selection

Fig. 11 compares the three visual models, and exposes a second unexplained deployment decision. EfficientNet-B0 outperforms ResNet-50 on accuracy (0.8208 against 0.7870), F1 (0.7967 against 0.7638), precision (0.7138 against 0.6724) and recall (0.9014 against 0.8840), while being five times smaller. ResNet-50 was nevertheless the model exported and deployed. No justification appears anywhere in the repository, and we mark it as *requires additional experiment*. The most plausible reconstruction is that ResNet-50 was trained and exported first and the pipeline was not revisited after EfficientNet-B0 was evaluated, but this is conjecture and we label it as such.

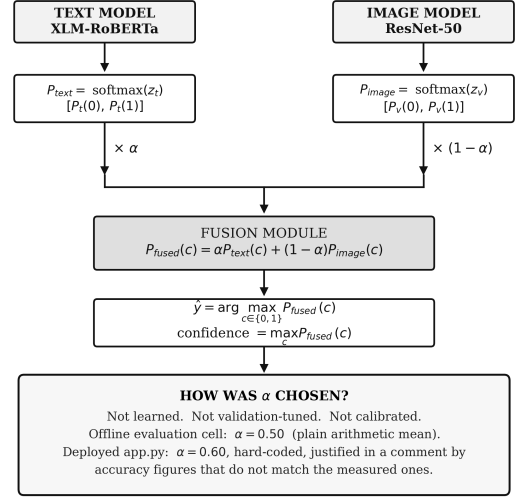
All three visual models share a signature: high recall (0.70–0.90) with low precision (0.67–0.71). Since the positive class is REAL, they systematically over-predict “real” on a corpus in which 61 % of items are fake. Section X argues that this is exactly what the grey-image failure path would produce.

VI. MULTIMODAL FUSION

A. Taxonomy

Following Baltrušaitis *et al.* [16], fusion strategies are conventionally grouped by the representational depth at which modalities meet. *Early* fusion concatenates raw or lightly processed inputs before any modality-specific encoding, so a single model learns cross-modal interaction from the ground up; it is expressive but demands large aligned corpora and is brittle when one modality is absent. *Intermediate* (feature-level) fusion encodes each modality separately, then merges the resulting embeddings inside a jointly trained network, allowing gradients to flow back into both encoders. *Late* (decision-level) fusion runs each modality to a complete prediction and combines only the outputs.

This project contains implementations of the latter two. **No early-fusion model was implemented**, and none is claimed.



Consequence: the only α for which a metric exists is 0.50, and that measurement used an untrained text branch (Sec. VIII-C).

Fig. 12. Weighted late-fusion mechanism of (2) and (3), together with a direct statement of how α was determined. As shown, the two branches meet only after each has produced a complete class distribution, so neither encoder ever observes the other modality.

B. Implemented late fusion

The deployed design combines the two class-probability vectors elementwise:

$$P_{\text{fused}}(c) = \alpha P_{\text{text}}(c | t) + (1 - \alpha) P_{\text{image}}(c | v) \quad (2)$$

with the decision and its confidence given by

$$\hat{y} = \arg \max_c P_{\text{fused}}(c), \quad \text{conf} = \max_c P_{\text{fused}}(c), \quad c \in \{0, 1\} \quad (3)$$

Since both operands are valid distributions and the weights are non-negative and sum to one, P_{fused} is itself a valid distribution, so (3) is well-formed. The module has no learnable parameters and requires no training pass, which is why Fig. 13 shows only two training loops. Fig. 12 renders the mechanism.

C. The choice of α

Here the implementation and the evaluation disagree. The offline evaluation cell computes $(\text{text_preds} + \text{image_preds}) / 2$, which is (2) with $\alpha = 0.50$. The deployed application computes $(\text{text_probs} * 0.6) + (\text{image_probs} * 0.4)$, which is (2) with $\alpha = 0.60$.

To answer the question directly and without embellishment: α was **not** learned, **not** tuned on a validation set, and **not** calibrated. It was chosen manually. The source comment accompanying the deployed value justifies it by asserting that the text model is more accurate than the image model, citing figures that do not match any measured result (Section XII-A, Table VII). The only α for which any metric exists in the repository is 0.50, and that measurement is itself invalid for the reason given in Section VIII-C. The deployed configuration has therefore never been evaluated.

We note in passing that with only two modalities, α is a single scalar on $[0, 1]$ and a proper sweep would cost eleven evaluation passes at a step of 0.1—a few minutes of compute. Its absence is not a resource limitation.

D. Implemented intermediate fusion

A second and architecturally different fusion model exists in the project, executed through the third-party codebase listed in Table I [35]. It is genuinely intermediate rather than late, and it is shown in Fig. 19. The text branch encodes the headline with a frozen *all-distilroberta-v1* sentence transformer [18], [19], yielding a 768-dimensional embedding pre-computed once and cached; the visual branch uses **ResNet-152**—not ResNet-50—with its final layer replaced by *Linear(2048, 300)*. Each branch is projected to 300 dimensions and passed through ReLU, the two are concatenated to 600 dimensions, and the joint vector flows through

$$h = \text{ReLU}(W_f[f_t; f_v] + b_f), \quad W_f \text{ of size } 512 \times 600 \quad (4)$$

followed by dropout at $p = 0.1$, a $512 \rightarrow 512$ ReLU layer and a $512 \rightarrow 2$ output projection trained with cross-entropy and Adam at 1×10^{-4} . Because the network is trained jointly and end-to-end, no per-modality probability is ever formed and (2) does not apply. Results appear in Section VIII-D and Table VI.

VII. EXPERIMENTAL SETUP

A. Hardware and software

All primary experiments ran on Google Colab with a single NVIDIA Tesla T4 GPU (16 GB), confirmed by the device query printed in the notebook. The stack is PyTorch [26] with torchvision, Hugging Face Transformers 4.35.0 [27] and scikit-learn [25]. The secondary experiment additionally used PyTorch Lightning and the *sentence-transformers* library [18]; one of its three runs executed on CPU after the GPU allocation lapsed, which is recorded explicitly in Table VI because it changes the wall-clock figures by an order of magnitude.

B. Optimisation settings

Textual models: AdamW [23], learning rate 2×10^{-5} , linear decay with zero warm-up steps, gradient-norm clipping at 1.0, batch size 16, sequence length 128, two epochs (three for XLM-RoBERTa). Visual models: Adam [24], learning rate 1×10^{-4} , constant schedule, no clipping, batch size 16, two epochs. Loss is unweighted cross-entropy in every case. Table IV collects these settings alongside the architectural parameters.

Two asymmetries in this protocol should be noted before the results are read. First, XLM-RoBERTa received 50% more epochs than the models it is compared against, which advantages it; that it still finished second to BERT-base is therefore mildly conservative evidence in BERT’s favour. Second, the visual learning rate is five times the textual one and is applied to a pre-trained transformer in the ViT case, where it is inappropriately large; this partially confounds the architecture comparison in Section V-E.

C. Metrics

Accuracy, precision, recall and F1 are computed with scikit-learn using *zero_division=0* and default binary averaging, so the positive class is label 1, which Section III-B established

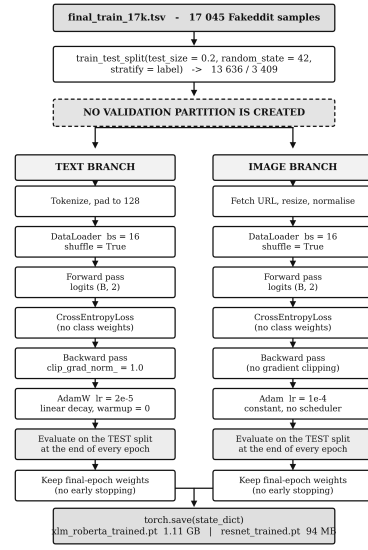


Fig. 13. Training pipeline for both branches. The two loops are independent and no third loop exists, because the late-fusion module of (2) has no parameters. Note the absence of a validation partition (Section III-E) and the differing optimiser settings between branches (Section VII-B).

to be REAL. Writing TP, FP, TN and FN with respect to that positive class,

$$\text{Acc} = \frac{\text{TP} + \text{TN}}{N}, \quad \text{Pr} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{Re} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad (5)$$

$$\text{F1} = \frac{2 \text{Pr} \text{Re}}{\text{Pr} + \text{Re}}$$

Every precision, recall and F1 in this paper therefore measures real-news retrieval. A reader interested in fake-news detection performance should mentally exchange the roles of the classes, or await the corrective experiment specified in Section XIV-C, which recommends reporting macro-averaged and per-class metrics instead. No confusion matrix was stored by any experiment, so the raw TP/FP/TN/FN counts are *not available in the current project artefacts*; the quantities in Table III are the only recorded evidence.

Fig. 13 and Fig. 14 set out the training and evaluation procedures respectively, in the exact form in which they were executed.

VIII. RESULTS

Table III is the principal result of this study. Every value in it was extracted from a stored notebook output cell and independently re-checked for internal consistency. Fig. 18 visualises the same data.

A. Textual results

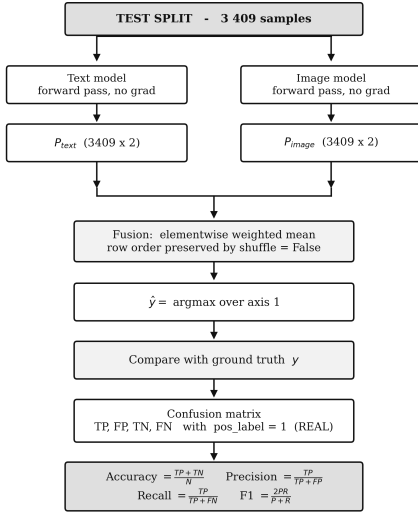
All four textual models clear the 0.6104 majority baseline decisively. The TF-IDF baseline reaches 0.7715, and the three transformers cluster tightly between 0.8483 and 0.8566—a spread of just 0.83 points, or roughly 28 test items. Contextual pre-training is therefore worth about 7.7 accuracy points over bag-of-bigrams on this corpus, while the choice among pre-trained transformers is worth almost nothing.

That flatness is itself informative. When three encoders with vocabularies of 30 522, 105 879 and 250 002 tokens, trained on Wikipedia and on CommonCrawl, converge to within one

TABLE III
MODELS EVALUATED IN THE STUDY

Model	Modality	Architecture	Pre-trained	Role	Acc.	Prec.	Rec.	F1
TF-IDF + LogReg	Text	Linear, 5k features, 1–2 gram	No	Baseline	0.7715	0.7431	0.6318	0.6829
mBERT	Text	12-layer transformer, $ V $ 105 879	Yes	Multiling. cand.	0.8542	0.7966	0.8404	0.8179
BERT-base-uncased	Text	12-layer transformer, $ V $ 30 522	Yes	Best text model	0.8566	0.8096	0.8261	0.8177
XLM-RoBERTa-base	Text	12-layer transformer, $ V $ 250 002	Yes	Deployed text	0.8483	0.8074	0.8020	0.8047
ViT-B/16	Image	12-layer ViT, 16×16 patches	Yes	Vision cand.	0.7580	0.6840	0.7041	0.6939
EfficientNet-B0	Image	16 MBConv blocks + SE	Yes	Best image model	0.8208	0.7138	0.9014	0.7967
ResNet-50	Image	50-layer bottleneck CNN	Yes	Deployed image	0.7870	0.6724	0.8840	0.7638
Late fusion ($\alpha = 0.5$)	Text+Image	Probability-level weighted mean	n/a	Reported system	0.7891 [†]	0.6759 [†]	0.8810 [†]	0.7650 [†]
Majority-class rule	—	Constant predictor	No	Trivial floor	0.6104	0.0000	0.0000	0.0000

All values are on the identical 3 409-sample stratified test split. The positive class for precision, recall and F1 is label 1 = REAL (Section III-B), so these three columns describe real-news retrieval. [†] The fusion row is invalid as a measurement of the intended system: the text branch used in that cell was randomly initialised (Section VIII-C, defect D1). The majority-class row is computed analytically from Table II, not measured.



Because the positive class is REAL, every precision, recall and F1 value in this study describes real-news retrieval, not fake-news detection.

Fig. 14. Evaluation pipeline. Row alignment between the textual and visual prediction matrices is preserved because both loaders are constructed with *shuffle=False* over index-aligned partitions, a property verified in Section III-A.

point of each other, the ceiling is being set by the data rather than by the model. Forty-two-character headlines carry a bounded amount of evidence, and all three encoders appear to be extracting most of it.

The precision-recall balance differs meaningfully across models even where accuracy does not. BERT-base is the most balanced (0.8096 precision against 0.8261 recall); mBERT leans towards recall (0.7966 against 0.8404); XLM-RoBERTa is almost exactly symmetric (0.8074 against 0.8020). The TF-IDF baseline is the outlier, with precision 0.7431 far exceeding recall 0.6318—it identifies real news conservatively, missing more than a third of it. Its F1 of 0.6829 is consequently 12 points below the transformers, a much larger gap than the accuracy comparison alone suggests. Table V records the per-epoch trajectories, plotted in Fig. 18(b).

B. Visual results

The visual models are weaker and more variable, spanning 0.7580 to 0.8208. EfficientNet-B0 leads, ResNet-50 follows

TABLE V
PER-EPOCH TEST ACCURACY AND F1

Model	Ep. 1 Acc / F1	Ep. 2 Acc / F1	Ep. 3 Acc / F1
mBERT	0.8366 / 0.8006	0.8542 / 0.8179	—
BERT-base	0.8530 / 0.8152	0.8566 / 0.8177	—
XLM-RoBERTa	0.8187 / 0.7341	0.8354 / 0.8039	0.8483 / 0.8047
ViT-B/16	0.7688 / 0.7372	0.7580 / 0.6939	—
EfficientNet-B0	0.8184 / 0.7897	0.8208 / 0.7967	—
ResNet-50	0.7832 / 0.7731	0.7870 / 0.7638	—

Bold marks the epoch whose weights were retained. ViT-B/16 is the only model that degraded; ResNet-50 gained accuracy while losing F1, indicating that the second epoch shifted its operating point rather than improving it.

at 0.7870, and ViT-B/16 trails at 0.7580—barely 14.8 points above the trivial baseline and 2 points below the TF-IDF text baseline. A model with 86.6M parameters and ImageNet pre-training failing to beat 5 000 TF-IDF features is a signal that something is wrong with the visual inputs, not merely with the visual architecture.

The shared precision-recall signature makes this concrete. All three models have recall for the REAL class between 0.70 and 0.90 but precision between 0.67 and 0.71. They over-predict REAL on a corpus that is 61 % FAKE. Section X links this directly to the grey-image failure path.

C. The fusion result, and why it is not a fusion result

The reported multimodal figure is 0.7891 accuracy, 0.6759 precision, 0.8810 recall and 0.7650 F1. It must not be interpreted as the performance of the intended system, for a reason established by source inspection rather than inference.

The fusion cell instantiates its text model as *AutoModelForSequenceClassification.from_pretrained('xlm-roberta-base', num_labels=2)* and immediately calls *.eval()*. The 1.11 GB fine-tuned checkpoint described in Section IV-E is never loaded. The cell’s own stored output confirms this, carrying the library warning that *classifier.dense* and *classifier.out_proj* were newly initialised. The “text” branch of this experiment was therefore a random projection of pre-trained features—an untrained classifier producing near-arbitrary probabilities.

The arithmetic corroborates the source reading. Compare the fusion row against ResNet-50 alone: accuracy 0.7891 against 0.7870, precision 0.6759 against 0.6724, recall 0.8810 against 0.8840, F1 0.7650 against 0.7638. Every metric agrees to

TABLE VI
INTERMEDIATE-FUSION RUNS (SECONDARY EXPERIMENT)

Run	Device	Train / Test	Batch	Epochs	Steps	Test loss	Test acc.
R1	CPU	150 / 50	8	1	13	0.6694	0.5625
R2	T4	1 200 / 300	8	3	294	0.7820	0.6125
R3	T4	4 000 / 1 000	16	5	825	0.8260	0.7020

Encoders: frozen all-distilroberta-v1 (768-d) and ResNet-152 (2048-d), each projected to 300-d, concatenated and classified through 600–512–512–2. Reported accuracy is the unweighted mean of per-batch accuracies, not a micro-average (see text). Only accuracy and loss were logged; no precision, recall, F1 or confusion matrix is available.

within 0.004. Under (2) with $\alpha = 0.5$, an uninformative P_{text} near $[0.5, 0.5]$ shifts P_{fused} towards the uniform distribution without changing which class is larger, so the arg max is inherited almost unchanged from the image branch. The observed near-identity is exactly the predicted consequence.

We therefore state the conclusion plainly. **This project has not measured the performance of its own multimodal system.** The correct experiment—loading the trained state dictionary before the prediction loop—is a one-line change and roughly ten minutes of T4 time. We decline to estimate what it would yield, because an estimate would be indistinguishable in the text from a measurement.

A secondary observation, worth recording for the corrected run: even a correctly executed late fusion may not improve on the text branch alone here. XLM-RoBERTa alone reaches 0.8483 while ResNet-50 reaches 0.7870, a gap of 6.1 points. Averaging a strong and a substantially weaker predictor with equal weight frequently lands between them. This is precisely why α should be selected on a validation split rather than fixed by hand, and why (2) should be evaluated across a sweep of α rather than at a single point.

D. Intermediate-fusion results

The secondary experiment (Fig. 19) trained the joint architecture of (4) three times at increasing scale. Table VI records all three runs, including the two that did not work well, because the progression is itself the finding.

Accuracy rises monotonically with data and compute—0.5625, 0.6125, 0.7020—which is the expected and reassuring pattern. But test *loss* rises in step, from 0.6694 to 0.7820 to 0.8260, while final training loss for R3 fell to 0.261. Accuracy improving while loss worsens is the classic signature of a model becoming more confident faster than it becomes more correct, i.e. growing miscalibration [29]. R3 is over-fitting in the probabilistic sense even as its arg max decisions improve. Any downstream use of these confidence scores—including display in a user interface—would be misleading.

Two measurement caveats bound what Table VI can support. First, the logged *test_acc* is $(1/B) \sum_b \text{acc}_b$ over batches rather than a micro-average over samples; with 43 test batches and a partial final batch, the two differ slightly and the reported figure is not exactly the overall accuracy. Second, image availability was the binding constraint: of 5 000 requested URLs, 1 679 failed and only 3 309 files reached disk, so R3’s nominal 4 000/1 000 split reduced to roughly 2 640 training and 688 evaluation samples after rows without images were dropped. This experiment at least *drops* unavailable images rather than substituting grey ones, making it methodologically cleaner than the primary pipeline on that specific point.

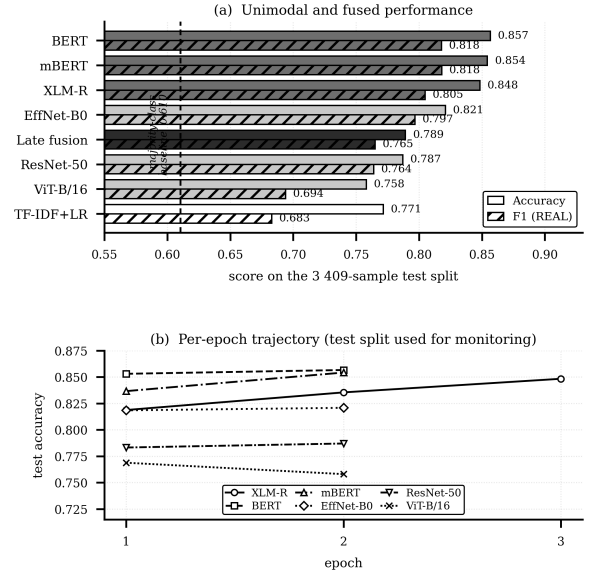


Fig. 18. (a) Accuracy and F1 for all evaluated models against the 0.6104 majority-class baseline; the fusion bar is shown for completeness but is invalid as a multimodal measurement (Section VIII-C). (b) Per-epoch test-accuracy trajectories from Table V. XLM-RoBERTa is still improving when training stops, whereas ViT-B/16 declines.

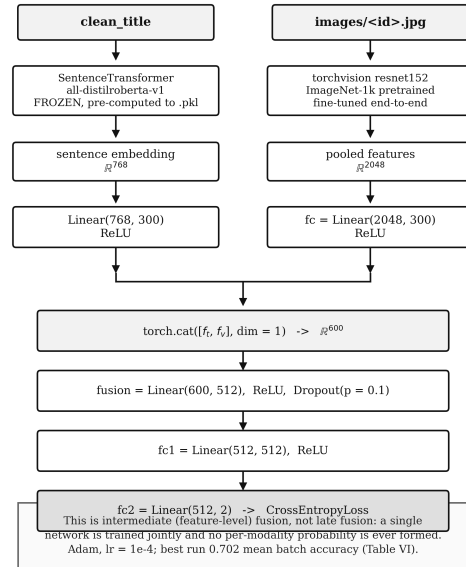


Fig. 19. Intermediate (feature-level) fusion architecture of the secondary experiment, corresponding to (4). Unlike Fig. 12, the two branches meet before any decision is formed and the whole network is trained jointly. Note that the vision backbone here is ResNet-152, not the ResNet-50 of the primary pipeline.

IX. MODEL COMPARISON

Table IV compares the architectures on the dimensions that explain the behaviour in Table III.

Three regularities emerge. **First, parameter count does not predict accuracy.** EfficientNet-B0, at 5.3 M parameters, beats ViT-B/16 at 86.6 M by 6.3 points and ResNet-50 at 25.6 M by 3.4. Architectural prior matters more than capacity at this data scale, exactly as the compound-scaling argument predicts [11].

Second, modality matters far more than architecture.

TABLE IV
ARCHITECTURE COMPARISON

Model	Params	Input	Representation	Classifier head	Training
TF-IDF + LR	≈ 5 k	Headline string	Sparse 5000-d tf-idf	Logistic regression	lbfgs, 1000 iter
mBERT	≈ 168 M	128 WordPiece ids	768-d [CLS] state	768–768–2, tanh	AdamW 2e–5, 2 ep.
BERT-base	≈ 110 M	128 WordPiece ids	768-d [CLS] state	768–768–2, tanh	AdamW 2e–5, 2 ep.
XLM-R base	≈ 278 M	128 SentencePiece ids	768-d <s> state	768–768–2, tanh	AdamW 2e–5, 3 ep.
ViT-B/16	≈ 86.6 M	196 patch tokens	768-d class token	Linear 768–2	Adam 1e–4, 2 ep.
EfficientNet-B0	≈ 5.3 M	$3 \times 224 \times 224$ tensor	1280-d pooled	Linear 1280–2	Adam 1e–4, 2 ep.
ResNet-50	≈ 25.6 M	$3 \times 224 \times 224$ tensor	2048-d pooled	Linear 2048–2	Adam 1e–4, 2 ep.
Joint fusion [35]	≈ 60 M+	768-d emb. + image	600-d concatenation	600–512–512–2	Adam 1e–4, 5 ep.

Parameter counts are the standard published figures for each checkpoint, not values measured in this project; the two-way heads change them by only a few thousand parameters. The joint-fusion row uses ResNet-152 and a frozen sentence encoder, so its trainable count is far below its total.

TABLE VIII
POSITIONING AGAINST PUBLISHED FAKEDDIT RESULTS

Work	Task	Modality	Reported	Comparable?
Nakamura et al. [3]	2-way	Text+image	≈ 0.89	No – full 1M corpus
Segura-Bedmar et al. [34]	6-way	Text+image CNN	≈ 0.87	No – different task
Segura-Bedmar et al. [34]	6-way	BERT text only	≈ 0.78	No – different task
This work	2-way	BERT text only	0.8566	Reference
This work	2-way	EfficientNet image	0.8208	Reference
This work	2-way	Late fusion	invalid	See Section VIII-C

Published figures are quoted from the cited works for orientation only. Because corpus size, label granularity and split protocol all differ, none of these rows constitutes a controlled comparison, and no claim of parity or superiority is made on their basis.

The worst text model (TF-IDF, 0.7715) outperforms the worst image model (ViT, 0.7580), and the best text model exceeds the best image model by 3.6 points. On this corpus the headline is simply more informative than the picture—partly because Fakeddit labels are derived from subreddits with strong lexical signatures, and partly because a third of the images may not have been retrieved at all.

Third, vocabulary size trades against everything else. XLM-R carries 2.5 times the parameters of BERT-base almost entirely in its embedding table, and scores 0.83 points lower on this monolingual English task. Those parameters buy cross-lingual coverage, which is worthless for the present benchmark and central to the roadmap of Section XIII.

X. ERROR ANALYSIS

No confusion matrix or per-sample prediction log was retained by any experiment, so conventional qualitative error analysis is *not available in the current project artefacts*. What follows is inference from the aggregate metrics and from the code, and we mark the boundary between the two explicitly.

A. The systematic visual bias (evidence-supported)

Every visual model exhibits recall substantially above precision for the REAL class. The mechanism is visible in the code. Suppose a fraction ρ of image fetches fail. Those samples become an identical grey tensor while retaining their original labels, which in a 61/39 fake/real corpus means the network is shown one constant input paired with contradictory targets. The loss-minimising response to a constant input with mixed labels is to predict the conditional class prior, and gradient descent will drive the grey-image response towards whichever class the head can most cheaply default to. Because these constant samples also populate the test set, they add an essentially random contribution to the reported metrics that is insensitive to the model.

We cannot compute ρ for the primary experiment because it was never logged. We can bound its plausible magnitude: the secondary experiment measured 33.6% on a comparable Fakeddit sample. Reddit preview CDN links expire, and the primary experiment additionally refetches every image every epoch under a five-second timeout on a shared Colab network. If ρ is anywhere near a third, then roughly 1150 of the 3409 test predictions were made on blank inputs, and the visual accuracies in Table III should be read as loose upper bounds on nothing in particular. **Measuring ρ is the single cheapest high-value experiment available to this project:** it requires one counter in the exception handler.

B. Shortcut learning (hypothesis, untested)

Fakeddit’s distant supervision makes subreddit identity a near-perfect proxy for the label. Real items come from r/nottheonion, r/upliftingnews and r/neutralnews; fake items come from r/psbattlartwork, r/fakehistoryporn and r/propagandaposters. These communities differ systematically in headline register and in image style—digital composites and vintage photographs against news photography and smartphone snapshots. A classifier can score highly by recognising the community rather than assessing veracity [28], [32].

The TF-IDF result is circumstantial evidence for this. A bag of 5000 bigrams with no semantic understanding whatsoever reaches 0.7715, which is 16.1 points above chance and within 8.5 points of BERT. That is a great deal of signal to be carried by surface lexis alone, and it is more consistent with register detection than with veracity assessment. We stress that this is a hypothesis.

The decisive test is a leave-one-subreddit-out evaluation: train on a subset of communities and test on held-out ones. If accuracy collapses, the models are recognising communities. This experiment was not performed, and it is the second corrective experiment recommended in Section XIV-C.

C. What the headline-only input cannot support

The textual input is a mean of 42 characters. There is no article body, no source domain, no publication date and no propagation context. Under these conditions a detector cannot verify a claim; at best it can recognise the stylistic markers of content drawn from communities that publish unverified material. Any deployment narrative implying factual verification therefore overstates what the artefact does, irrespective of the accuracy achieved.

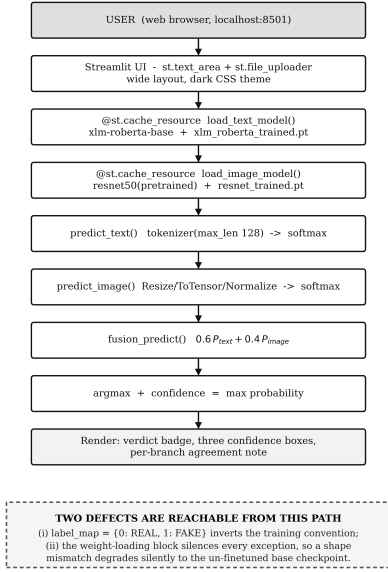


Fig. 15. Streamlit deployment path. Preprocessing is consistent with training, which is a real strength; the two defects annotated at the foot of the figure are analysed in Sections XI-A and XI-B.

XI. DEPLOYMENT

The system is deployed as a 459-line Streamlit application, shown in Fig. 15. The user supplies a headline and uploads an image; two `@st.cache_resource` loaders instantiate the encoders once per session and attach the exported state dictionaries; `predict_text` and `predict_image` produce softmax vectors; `fusion_predict` applies (2) with $\alpha = 0.60$; and the interface renders a verdict badge, three confidence boxes and a short note on whether the branches agreed. Preprocessing inside the application matches training exactly—the same 128-token cap and the same ImageNet normalisation constants—which is a genuine strength, since train/serve preprocessing skew is a common and silent source of degradation.

Two defects are nevertheless reachable from this path, and both are in the application rather than in the models.

A. Inverted label mapping

The application defines `label_map = {0: "REAL NEWS", 1: "FAKE NEWS"}`. Section III-B established from the data that label 1 is REAL and label 0 is FAKE. The mapping is inverted, and because it is applied uniformly to the text, image and fusion verdicts, the interface reports the exact opposite of every prediction the models make. The same incorrect convention appears in the header of the *TruthLens* notebook, which suggests a shared misreading rather than a typing slip. The fix is a two-character change; the impact until then is total.

B. Silent fallback to un-finetuned weights

Both loaders wrap `load_state_dict` in a bare `except` that falls back to `strict=False` and then to the un-finetuned base checkpoint, recording only an internal status string. A shape mismatch, a corrupt download or a missing file therefore degrades the application to random-head inference without any user-visible signal. Given that defect D1 of Section VIII-C is the same failure occurring in the evaluation code, this is not

TABLE VII
DOCUMENTED VERSUS MEASURED ACCURACY

Model	Documented	Measured	Δ (pp)	Status
TF-IDF + LogReg	0.7715	0.7715	0.00	Agrees
mBERT	0.8120	0.8542	-4.22	Contradicted
BERT-base	0.8250	0.8566	-3.16	Contradicted
XLM-RoBERTa	0.8390	0.8483	-0.93	Contradicted
ViT-B/16	0.7785	0.7580	+2.05	Contradicted
EfficientNet-B0	0.8128	0.8208	-0.80	Contradicted
ResNet-50	0.8046	0.7870	+1.76	Contradicted
Late fusion	0.8049	0.7891	+1.58	Both invalid [‡]

Δ is documented minus measured, in percentage points. [‡] The measured fusion value is itself invalid as a system measurement because of defect D1. All values in this paper's other tables are the measured column.

a hypothetical risk: the project has already been bitten once by an unloaded checkpoint that produced plausible-looking numbers.

The remedy in both places is to let the exception propagate. Silent recovery from a failed weight load converts a loud, immediate error into a quiet, permanent one.

XII. LIMITATIONS AND VERIFICATION REPORT

This section discharges the verification obligation. Every figure in the project's own documentation was cross-checked against the notebook cell that produced it. Where the two disagree, the notebook output is authoritative and the discrepancy is recorded rather than reconciled.

A. Documentation-measurement discrepancy (D4)

Table VII compares the accuracies stated in the repository's documentation with those recovered from the executed notebooks. Only the TF-IDF baseline agrees. Six of the remaining seven differ by more than a point, in both directions, so the pattern is not a systematic offset such as a different split or a different epoch. The documentation additionally reports a per-epoch XLM-RoBERTa trajectory of 0.7979 / 0.8181 / 0.8390 that appears in no notebook; the measured trajectory is 0.8187 / 0.8354 / 0.8483. It also states a paired *t*-test with $p = 0.087$, but no statistical test is implemented anywhere in the repository. These values must be regarded as unsupported.

We draw attention to one detail with real consequences. Because the documented figures happen to place the image models above their measured values and the text models below, they make late fusion look nearly free (0.8049 documented against 0.8390 for text). The measured values tell a different story: fusion at 0.7891 sits 5.9 points below XLM-RoBERTa alone. The incorrect numbers thus supported a design decision—the 60/40 weighting—that the correct numbers do not.

B. Enumerated defects

Table IX lists each defect, its evidence, its effect and its remedy. We rank D1 as most severe because it invalidates the project's central claim, D3 next because it renders the demonstration artefact actively misleading, and the remainder as significant but bounded.

TABLE IX
RESEARCH-INTEGRITY VERIFICATION REPORT

ID	Defect	Evidence	Effect on conclusions	Remedy
D1	Fusion evaluated with a randomly initialised text head	Source: checkpoint never loaded; stored HF warning; fusion metrics match ResNet-50 to within 0.004	Multimodal claim unsupported	Load state dict; rerun (≈ 10 min)
D2	$\alpha = 0.50$ offline but 0.60 deployed	Notebook divides by 2; app.py uses 0.6/0.4	Deployed configuration never evaluated	Sweep α on a validation split
D3	Label map inverted in the application	Cross-tab of 2-way vs 6-way label vs subreddit over all 17 045 rows	Demo reports the opposite of every prediction	Swap the two dictionary values
D4	Documented accuracies contradict notebook outputs	Table VII; 7 of 8 models disagree	Documentation unusable as evidence	Regenerate from results JSON
D5	No validation partition	Single train_test_split; test set monitored each epoch	Reported accuracies optimistic	Three-way split 70/15/15
D6	Failed image fetches become grey tensors with original labels	Exception handler in ImageDataset; 33.6 % failure measured on a comparable sample	Visual metrics unreliable at stated precision	Log ρ ; drop failed rows
D7	Positive class is REAL, not FAKE	scikit-learn default pos_label = 1; Section III-B	P/R/F1 describe the wrong class for the stated aim	Report per-class and macro metrics
D8	Distant supervision by subreddit	Table II source communities; TF-IDF reaches 0.7715	Possible register detection, not veracity	Leave-one-subreddit-out evaluation

Every entry in the evidence column was verified directly against repository source or against a stored notebook output cell. No defect in this table is inferred solely from a metric anomaly.

C. Further limitations

Beyond the enumerated defects, four constraints bound this study. The corpus is a 17 045-sample extract of a corpus exceeding one million, so absolute accuracies are not comparable with published Fakeddit results (Table VIII). Training budgets of two to three epochs mean at least XLM-RoBERTa is under-trained. Every model was run once with a single seed, so no variance estimate exists and differences of under one point should not be interpreted. Finally, the entire study is English-only and Reddit-only; nothing here establishes transfer to Bangla, to professionally published news, or to any other platform.

XIII. TOWARDS BANGLA MULTIMODAL DETECTION

Nothing in this section has been implemented or evaluated. It is future work, and no result is claimed for it. The project’s longer-term ambition is to extend multimodal misinformation detection to Bangla, a language with more than 200 million speakers and negligible coverage in this literature. Fig. 16 sets out the roadmap and Fig. 17 sketches a candidate architecture.

A. The binding constraint is the corpus

It is tempting to frame a Bangla extension as a modelling problem. It is not. BanFakeNews [20] provides roughly 50 000 annotated Bangla articles, but it is text-only; to our knowledge no publicly available Bangla dataset pairs news text with the accompanying image and a veracity label. Every architecture in Fig. 17 could be assembled in an afternoon from existing

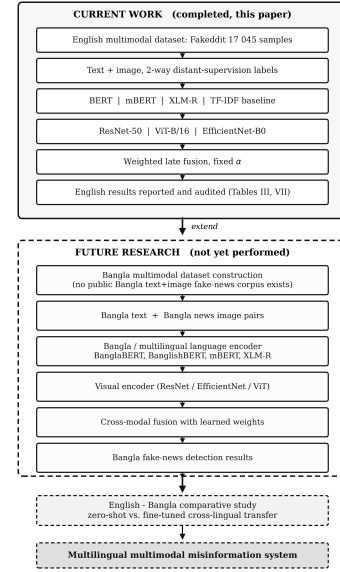


Fig. 16. Research roadmap from the completed English system to a multi-lingual multimodal one. The solid enclosure is work performed and audited in this paper; everything below the dashed boundary is future work for which no result is claimed.

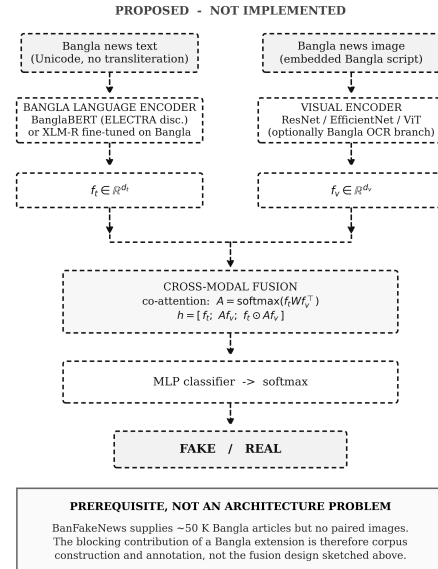


Fig. 17. Candidate Bangla multimodal architecture with cross-modal co-attention. Every element is drawn dashed to signal that it is proposed rather than implemented. As argued in Section XIII-A, the obstacle is the missing paired corpus, not this design.

components. None of them can be trained. The blocking contribution of a Bangla multimodal project is therefore **dataset construction and annotation**, which is slower, less glamorous and considerably more valuable than another fusion variant.

A realistic construction plan would harvest from Bangladeshi fact-checking organisations and mainstream outlets, retain the image alongside the claim, and annotate with multiple raters and a reported inter-annotator agreement statistic. Even 5 000 well-annotated paired samples would be a publishable resource contribution, and would be worth more to the field than any accuracy number this project could produce on Fakeddit.

B. Candidate encoders

For the language branch, BanglaBERT [21] is the strongest published option: an ELECTRA-style discriminator pre-trained on 27.5GB of Bangla text that outperforms both mBERT and XLM-R (large) on Bangla understanding tasks while being markedly cheaper to fine-tune. XLM-RoBERTa [7]—already fine-tuned in this study—is the natural cross-lingual baseline, and mBERT [8] permits zero-shot transfer from the English models trained here. IndicBERT [22] is a further multilingual alternative. For the visual branch, the encoders of Section V transfer directly, since ImageNet features are not language-specific; an auxiliary Bangla OCR stream would be a sensible addition, because a great deal of Bangla misinformation circulates as text rendered into an image, which a purely visual encoder cannot read.

C. The comparative study that would follow

The scientifically interesting question is not whether a Bangla detector can be built, but how cross-lingual transfer behaves for this task. A three-arm comparison would answer it: zero-shot application of the English-trained mBERT and XLM-R models to Bangla; those same models fine-tuned on Bangla data; and BanglaBERT trained on Bangla from its own pre-training. The gap between arms one and two quantifies how much Bangla supervision is actually needed, and the gap between two and three quantifies the value of monolingual pre-training. Because the present study already produced fine-tuned mBERT and XLM-R checkpoints on English Fakeddit, arm one is nearly free once a Bangla evaluation set exists.

XIV. DISCUSSION AND RESEARCH GAP

A. What prior work has settled, and what it has not

The literature has established that multimodal detectors can outperform unimodal ones on curated benchmarks [13]–[15], that transformer encoders dominate feature-engineered text pipelines [5], [7], and that large paired corpora such as Fakeddit [3] make supervised training feasible at scale. What it has not settled is more interesting. Generalisation across events, platforms and time remains weak; EANN’s event-adversarial design [13] exists precisely because detectors memorise events. Benchmarks with distantly supervised labels reward register recognition over veracity assessment [28], [32]. Reported multimodal gains are frequently small relative to the strongest unimodal branch and are rarely accompanied by an ablation isolating the contribution of each modality. And low-resource languages, Bangla among them, are almost entirely absent from the multimodal literature [20].

B. What this project does and does not demonstrate

It demonstrates four things. A controlled seven-model comparison under a fixed data and optimisation budget, which is more disciplined than much of the published work it draws on. A competitive text result: 0.8566 accuracy for BERT-base on binary Fakeddit, 24.6 points above the majority baseline. A working end-to-end deployment with train/serve preprocessing parity. And, unusually, a documented audit of its own measurement chain.

It does not demonstrate a multimodal gain (defect D1). It does not demonstrate that the deployed 60/40 weighting is

sound (D2). It does not demonstrate reliable visual classification, because the image-retrieval failure rate is unmeasured (D6). It does not demonstrate veracity assessment as distinct from community-register recognition (D8). And it does not demonstrate anything whatsoever about Bangla.

C. The corrective experiment set

The gap between what this project built and what it proved is narrow and closable. In descending order of value per unit effort:

- 1) **Rerun fusion with the trained checkpoint.** One line, roughly ten minutes on a T4. Converts the central claim from unsupported to measured. Without it, no multimodal statement in this project stands.
- 2) **Instrument the image fetch.** One counter in the exception handler. Establishes whether the visual results in Table III mean anything.
- 3) **Correct the deployed label map.** Two characters. Until then the demonstration artefact reports the inverse of its own predictions.
- 4) **Sweep α on a proper validation split.** Eleven evaluation passes at step 0.1. Replaces a hand-picked constant with a tuned one and yields the α -versus-accuracy curve that a reviewer will expect.
- 5) **Report per-class and macro-averaged metrics with a confusion matrix.** Makes fake-news detection performance visible, which is the stated aim.
- 6) **Run leave-one-subreddit-out evaluation.** The one experiment that would distinguish veracity assessment from register recognition, and by far the most publishable single addition to this work.

The first five are hours of work in total. The sixth is a day. Together they would convert a competent engineering project into a defensible empirical one.

D. The research gap, stated directly

The gap this project sits within is not the absence of a multimodal architecture for misinformation detection—several exist and work. It is the absence of trustworthy evaluation of such architectures on distantly supervised benchmarks, and the absence of any multimodal resource for Bangla. The present study contributes to the first by demonstrating, on itself, how easily a plausible multimodal number can arise from an unloaded checkpoint. It contributes to the second only by identifying precisely what is missing.

XV. CONCLUSION

We benchmarked four textual and three visual classifiers on an identical 17 045-sample Fakeddit subset under a fixed budget, combined the deployed branches with weighted late fusion, and audited every resulting number against the code that produced it. BERT-base-uncased achieved the highest accuracy at 0.8566, with mBERT at 0.8542 and XLM-RoBERTa at 0.8483; EfficientNet-B0 led the visual models at 0.8208 against ResNet-50’s 0.7870 and ViT-B/16’s 0.7580. Parameter count predicted accuracy poorly, and modality mattered more than architecture.

The audit is the part we would ask a reader to remember. The project’s headline multimodal figure was produced by fusing a trained image branch with a randomly initialised text head, and it is consequently a measurement of the image branch alone; the deployed fusion weight has never been evaluated; and the deployed interface inverts the dataset’s label convention. None of these is exotic. Each is the kind of defect that survives review precisely because the resulting numbers look reasonable. We report them because a study that cannot locate its own errors provides no basis for trusting the errors it did not look for.

The system therefore stands as a competent unimodal benchmark and a functioning deployment, and not yet as a demonstration of multimodal benefit. Six corrective experiments, five of them costing hours rather than days, would close that gap. Beyond them, the substantive opportunity lies in Bangla, where the binding constraint is the construction of a paired text-image misinformation corpus rather than the design of a model to consume it.

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Reproducibility statement. Every metric reported in this paper was extracted from a stored output cell of an executed notebook in the project repository. Values that could not be recovered are marked *not available in the current project artefacts*; experiments that were never run are marked *requires additional experiment*. No metric in this paper was estimated, interpolated or reproduced from documentation.